

Concurrent Programming TDA384/DIT392

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(Exam set by N. Piterman and G. Schneider, based on the course given in
Aug-Oct 2023 and Jan-Mar 2024)

Material permitted during the exam (hjälpmedel):

Two textbooks; four sheets of A4 paper with notes (single or double-sided);
English dictionary (no smart phones allowed).

Grading: You can score a maximum of 70 points. Exam grades are: between 28–41 (3), between 42–55 (4), 56 or more (5).

Passing the course requires passing the exam and passing the labs. The overall grade for the course is determined as follows: between 40–59 (3), between 60–79 (4), 80 or more (5).

The exam **results** will be available in Ladok within 15 *working* days after the exam's date.

Instructions and rules:

- Please write your answers clearly and legibly: unnecessarily complicated solutions will lose points, and answers that cannot be read will receive no points!
- Justify your answers, and clearly state any assumptions that your solutions may depend on for correctness.
- Answer each question on a new page. Glance through the whole paper first; five questions, numbered Q1 through Q5. Do not spend more time on any question or part than justified by the points it carries.
- Be precise. In your answers, try to use the programming notation and syntax used in the questions. You can also use pseudo-code, *provided* the meaning is precise and clear. If need be, explain your notation.

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Q1 (12p). In what follows you have 4 sub-questions (Parts a to d) concerning different topics seen in the course. Each part a multiple-choice question. The grading for each part is as follows:

- Correct: 3 points;
- No answer: 0 points;
- Incorrect: -1 point (negative).

The total number of points will be done by summing all the points for each part capped from below at 0. That is, minus points are *local* to this question but do not affect other questions.

We did not apply the -1 penalty for incorrect answers in this question.

NOTE: For each sub-question, there is exactly one option that is correct.

(Part a). (3p)

Consider the following program:

```
6      private static volatile boolean flag;
7
8      public void function1() {
9
10         final Thread t1 = new Thread(() -> {
11             for (int i=0 ; i<10 ; i++) {
12                 flag = true;
13             }
14         });
15
16         final Thread t2 = new Thread(() -> {
17             for (int i=0 ; i<10 ; i++) {
18                 flag = true;
19             }
20         });
21
22         t1.start();
23         t2.start();
24     }
```

- a) It has a data race.
- b) It does not have a data race but has a race condition.
- c) It has both a data race and a race condition.
- d) It is fine.

(Part b). (3p)

Consider the following program:

```
34     private volatile int counter;
35
36     public void function2() {
37         final Thread t3 = new Thread(() -> {
38             for (int i=0 ; i<10 ; i++) {
39                 int cnt = counter;
40                 counter = cnt + 1;
41             }
42         });
43
44         final Thread t4 = new Thread(() -> {
45             for (int i=0 ; i<10 ; i++) {
46                 int cnt = counter;
47                 counter = cnt + 1;
48             }
49         });
50
51         t3.start();
52         t4.start();
53     }
```

- a) It has a data race.
- b) It does not have a data race but has a race condition.
- c) It has both a data race and a race condition.
- d) It is fine.

(Part c). (3p)

- a) In Erlang one cannot have race conditions.
- b) In Erlang one cannot have data races.
- c) In Erlang neither data races nor race conditions are possible.
- d) In Erlang both data races and race conditions are possible.

(Part d). (3p)

Consider the following usage of monitor:

```
24 |         l.lock();
25 |     try {
26 |         if (check_something) {
27 |             cond.await();
28 |         }
29 |         // do something
30 |     }
31 |     finally {
32 |         l.unlock();
33 |     }
```

- a) This is a good example of using a condition in a monitor.
- b) This is a good example of using a condition in a monitor under signal-and-wait.
- c) This is a good example of using a condition in a monitor under signal-and-continue.
- d) This is not the right way to use a condition in a monitor.

Q2 (18p). In what follows you will get 4 assertions concerning topics taught in the course. For each assertion, you need to say whether it is correct (true) or not (false). You need to justify your answer in each case (an answer without a justification will not be granted full points).

(Part a). (5p) You are using the software transactional memory that was taught in class to manage a set of variables (see below). Every time you finish a transaction you send all the variables you checked out back and their version numbers are updated.

```
stm(Storage, {pull, Name}) ->
  case dict:is_key(Name, Storage) of
    true ->
      {reply, Storage,
       dict:fetch(Name, Storage)};
    false ->
      {reply, Storage, not-found}
  end;

stm(Storage, {push, Vars}) ->
  case try-push(Vars, Storage) of
    {success, NewStorage} ->
      {reply, NewStorage, success};
    fail ->
      {reply, Storage, fail}
  end.

try-push([], Storage) ->
  {success, Storage};
try-push([Var = #var{name = Name, version = Version} | Vars], Storage) ->
  case dict:find(Name, Storage) of
    {ok, #var{version = Version}} ->
      try-push(Vars,
                dict:store(Name, Var#var{version = Version + 1},
                           Storage));
    _ -> fail
  end.
```

A colleague tells you that this approach is inefficient as it updates the version numbers of variables that have not changed. They suggest to send back only the variables that were changed and that those that were not changed do not need to be pushed back. This is a bad suggestion.

(Part b). (5p)

You are struggling to create a pair of processes in Erlang that exchange messages with each other. A colleague suggests that you could use the following approach. Use the code below to create several channels (say a2b and b2a). Then, create the processes that you want to talk to each other (say a and b) and give them the channels that were created in advance.

```

1  -module(channel).
2  -export([start/1,get_from_channel/1,send_through_channel/2]).
3
4  start(Channel) ->
5  |   Pid = spawn(fun channel_fun/0),
6  |   catch unregister(Channel),
7  |   register(Channel, Pid).
8
9  channel_fun() ->
10 |   receive
11 |   |   {get, From1, Ref1 } ->
12 |   |   |   receive
13 |   |   |   |   { put, From2, Ref2, Payload } ->
14 |   |   |   |   |   From1 ! { reply, Ref1, Payload },
15 |   |   |   |   |   From2 ! { ack, Ref2 }
16 |   |   |   end,
17 |   |   channel_fun
18 |   end.
19
20 send_through_channel(Channel, Data) ->
21 |   Ref = make_ref(),
22 |   Channel ! { put, self(), Ref, Data},
23 |   receive
24 |   |   { ack, ref } -> ok
25 |   end.
26
27 get_from_channel(Channel) ->
28 |   Ref = make_ref(),
29 |   Channel ! {get, self(), Ref},
30 |   receive
31 |   |   {reply, Ref, Payload} -> Payload
32 |   end.

```

You go ahead and use this code. Just to be on the safe side, you use several channels for each process (so a2b1, a2b2, a2b3 and so on). Your colleague is alarmed and tells you that it is no longer the case that if process a sends two messages to process b they arrive in the order that they were sent. Your colleague is wrong.

(Part c). (5)

Your colleague then supplies you with this example of how to create two processes that communicate with each other:

```
28  compute() ->
29      ok.
30
31  send_to_other(Other,Data) ->
32      Other ! { Data, self() }.
33
34  get_from_other(Other) ->
35      receive
36      |   { Data, Other } -> Data
37  end.
38
39  connected_process(Other) ->
40      Data = compute(),
41      send_to_other(Other,Data),
42      NewData = get_from_other(Other),
43      connected_process(Other).
44
45  unconnected_process() ->
46      receive
47      |   Pid -> connected_process(Pid)
48  end.
49
50  main() ->
51      First = spawn(fun unconnected_process/0),
52      Second = spawn(fun () -> connected_process(First) end),
53      First ! Second.
54
```

This indeed solves the problem (and is a better and simpler option).

(Part d). (3p)

Consider the following usage of lock and semaphore:

```
10      public void useSemaphore() throws InterruptedException {
11          l.lock();
12          if (s.availablePermits() > 0) {
13              s.acquire();
14          } else {
15              l.unlock();
16              s.acquire();
17          }
18      }
```

The way the lock and semaphore are used is dangerous and could lead to a deadlock.

Q3 (11p). Here is an skeleton of an implementation of an algorithm implemented using fork/join parallelism and a workers pool. We are interested in the number of tasks that are going to be created by a run of this algorithm.

The algorithm starts from a set of elements (**pairs**). Then, for every element of the set (line 26), it creates a smaller set with the rest of the elements (lines 27-28), and then creates and forks a recursive task for each of them (lines 33-34). Before making a final computation in **summarize()** (line 43), the algorithms joins all the recursive tasks that were created (lines 40-42).

```

11  static class StreettSolverTask extends RecursiveTask<Integer> {
12      private Integer something_important;
13      private HashSet<Integer> pairs;
14
15      public StreettSolverTask(HashSet<Integer> pairs, Integer something_important) {
16          this.pairs = pairs;
17          this.something_important = something_important;
18      }
19
20      protected Integer compute() {
21          if (pairs.size() <= 1) {
22              return something_important;
23          }
24
25          List<StreettSolverTask> l = new ArrayList<StreettSolverTask>();
26          for (Integer pair: pairs) {
27              HashSet<Integer> other_pairs = new HashSet<Integer>(pairs);
28              other_pairs.remove(pair);
29
30              // Do some important computation with
31              // the value of something important
32
33              StreettSolverTask t = new StreettSolverTask(other_pairs,something_important);
34              t.fork();
35
36              l.add(t);
37          }
38
39          List<Integer> results = new ArrayList<Integer>();
40          for (StreettSolverTask t : l) {
41              results.add(t.join());
42          }
43          return summarize(results);
44      }
45
46  > private Integer summarize(List<Integer> list) {-
49  }

```

(Part a). (6p)

If the initial task is created with a set of size 4, how many recursive tasks are going to be created (including the initial one). Explain your answer. It is OK to give a calculation without a final number.

(Part b). (4p)

Discuss the potential advantages and disadvantages of running this parallel implementation compared with a direct application of Threads rather than using a pool and compared with a sequential implementation.

Q4 (15p). Here is an Erlang implementation of a monitor with a single condition:

```
1  -module(monitor).
2  -export([start/0,lock/1,unlock/1,wait/1,signal/1]).
3
4  start() ->
5  |   spawn(fun () -> loop([]) end).
6
7  loop(Blocked) ->
8  |   receive
9  |   |   { lock, Owner } ->
10 |   |   |   Owner ! { ok, self() },
11 |   |   |   loop(Owner,Blocked)
12 |   |   end.
13
14  loop(Owner, Blocked) ->
15  |   receive
16  |   |   { unlock, Owner } -> loop(Blocked);
17  |   |   { wait, Owner } -> loop([Owner | Blocked]);
18  |   |   { signal, Owner } ->
19  |   |   |   case Blocked of
20  |   |   |   |   [ Pid | Tail ] -> Pid ! { continue, self() };
21  |   |   |   |   Tail -> ok
22  |   |   |   end,
23  |   |   |   loop(Owner,Tail)
24  |   |   end.
25
26  lock(Monitor) ->
27  |   Monitor ! { lock, self() },
28  |   receive
29  |   |   { ok, Monitor } -> ok
30  |   |   end.
31
32  unlock(Monitor) ->
33  |   Monitor ! { unlock, self() }.
34
35  wait(Monitor) ->
36  |   Monitor ! { wait, self() },
37  |   receive
38  |   |   { continue, Monitor } -> ok
39  |   |   end,
40  |   |   lock(Monitor).
41
42  signal(Monitor) ->
43  |   Monitor ! { signal, self() }.
```

(Part a). (3p)

Is the signalling policy implemented by the monitor singal-and-wait or signal-and-continue?

v **(Part b). (8p)**

Modify the implementation so that it implements the other signaling policy.

(Part c). (4p)

Going back to the original implementation, sketch how you would modify it to have multiple conditions. There is no need to implement your solution but only explain how you would go about it.

You do not have to solve part b to score points in part c.

Q5 (14p). Here are programs for controlling crossings of a bridge between the two peaks of the Kilimanjaro. The bridge cannot hold more than two people together. The programs try to be fair by not allowing two people to cross in one direction if someone is waiting to cross in the other direction. For each solution, identify whether the solution works as promised. If it works, explain why. If it does not work, show its problems.

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(Part a). (7p) Here is a monitor based solution (note the use of monitor keyword).

```

5      monitor class Kilimanjaro1 {
6
7          int waitL=0,waitR=0,crossingL=0,crossingR=0;
8          Condition goL,goR;
9
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11
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```

(Part b). (7p) Here is a semaphore based solution.

```

3  public class Kilimanjaro2 {
4      Semaphore waitL;
5      Semaphore waitR;
6      Semaphore crossingL;
7      Semaphore crossingR;
8
9      void Kilomanjaro2() {
10         waitL = new Semaphore(permits:0);
11         waitR = new Semaphore(permits:0);
12         crossingL = new Semaphore(permits:1);
13         crossingR = new Semaphore(permits:1);
14     }
15
16     public void registerL() {
17         waitL.release();
18     }
19
20     public void registerR() {
21         waitR.release();
22     }
23
24     public void enterL() throws InterruptedException {
25         if (!crossingL.tryAcquire()) {
26             if (waitR.tryAcquire()) {
27                 // someone waiting on the other side
28                 waitR.release();
29                 crossingL.acquire();
30             } else {
31                 crossingR.acquire();
32             }
33         }
34         // Acquired an can cross the bridge
35         waitL.acquire();
36     }
37
38     public void enterR() throws InterruptedException {
39         if (!crossingR.tryAcquire()) {
40             if (waitL.tryAcquire()) {
41                 // someone waiting on the other side
42                 waitL.release();
43                 crossingR.acquire();
44             } else {
45                 crossingL.acquire();
46             }
47         }
48         // Acquired an can cross the bridge
49         waitR.acquire();
50     }
51
52     public void exitR() {
53         crossingR.release();
54     }
55
56     public void exitL() {
57         crossingL.release();
58     }

```